Discriminant Analysis

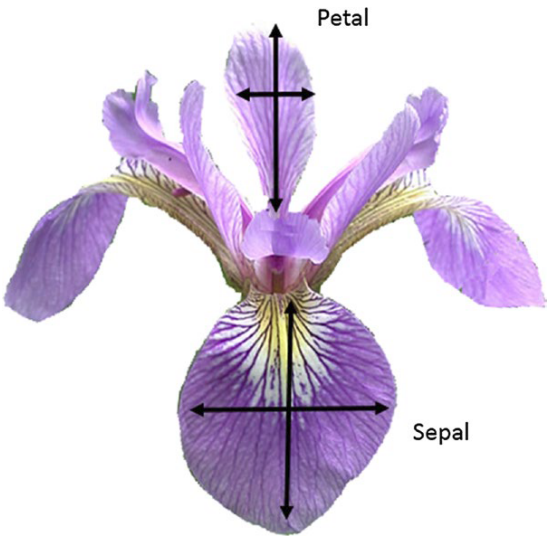
Ngoc Hoang Luong

University of Information Technology (UIT), VNU-HCM

October 26, 2026

Motivation

The **Iris dataset** is a collection of 150 labeled examples of Iris flowers, 50 of each type, described by 4 features: sepal length, sepal width, petal length, petal width.

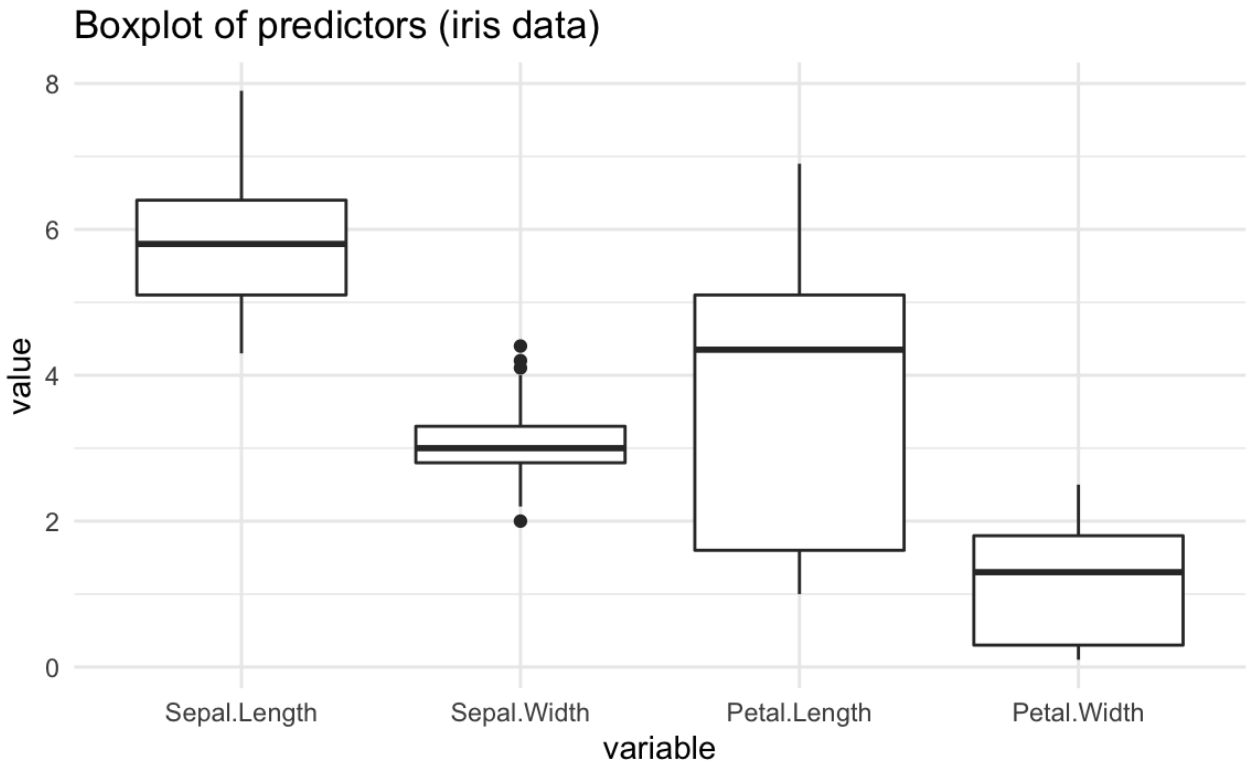


index	sl	sw	pl	pw	label
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
	...				
50	7.0	3.2	4.7	1.4	Versicolor
	...				
149	5.9	3.0	5.1	1.8	Virginica

Figure: A subset of the Iris design matrix.

Motivation

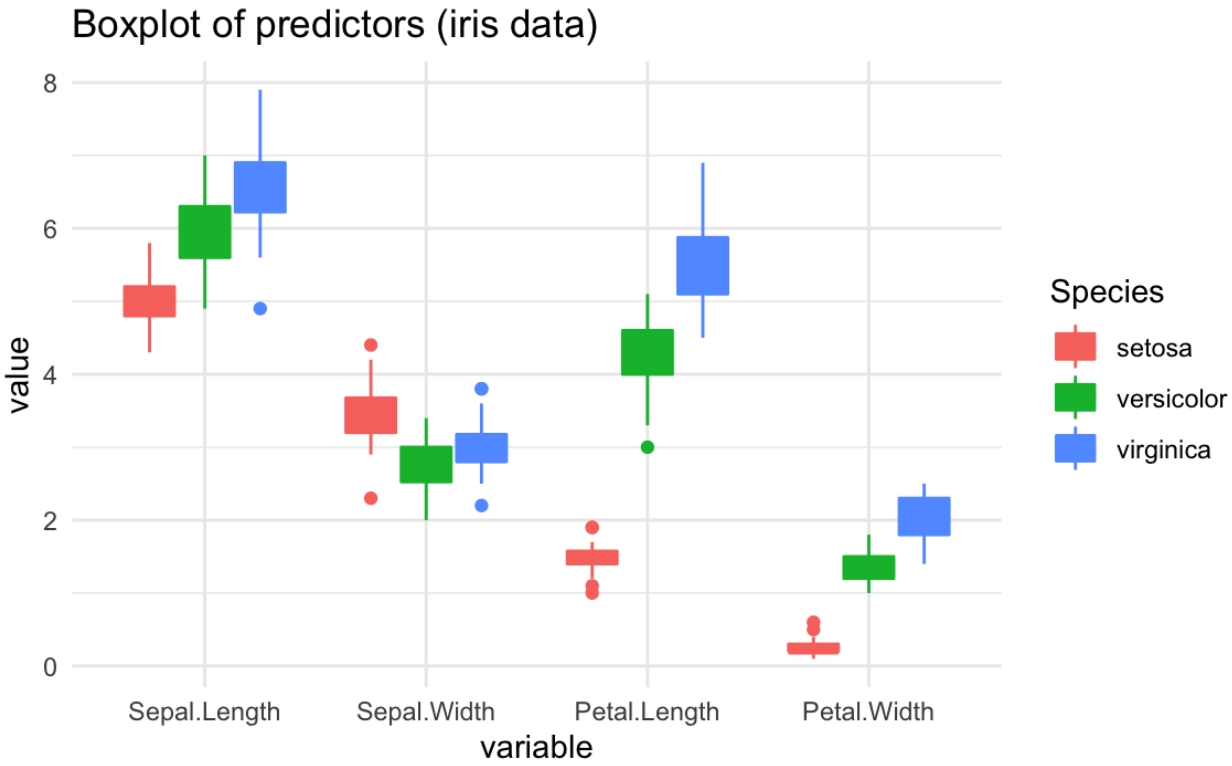
- Let's look at the distribution of the four input variables without making distinction between species of iris flowers:



- All predictors have different ranges, as well as different types of distributions.

Motivation

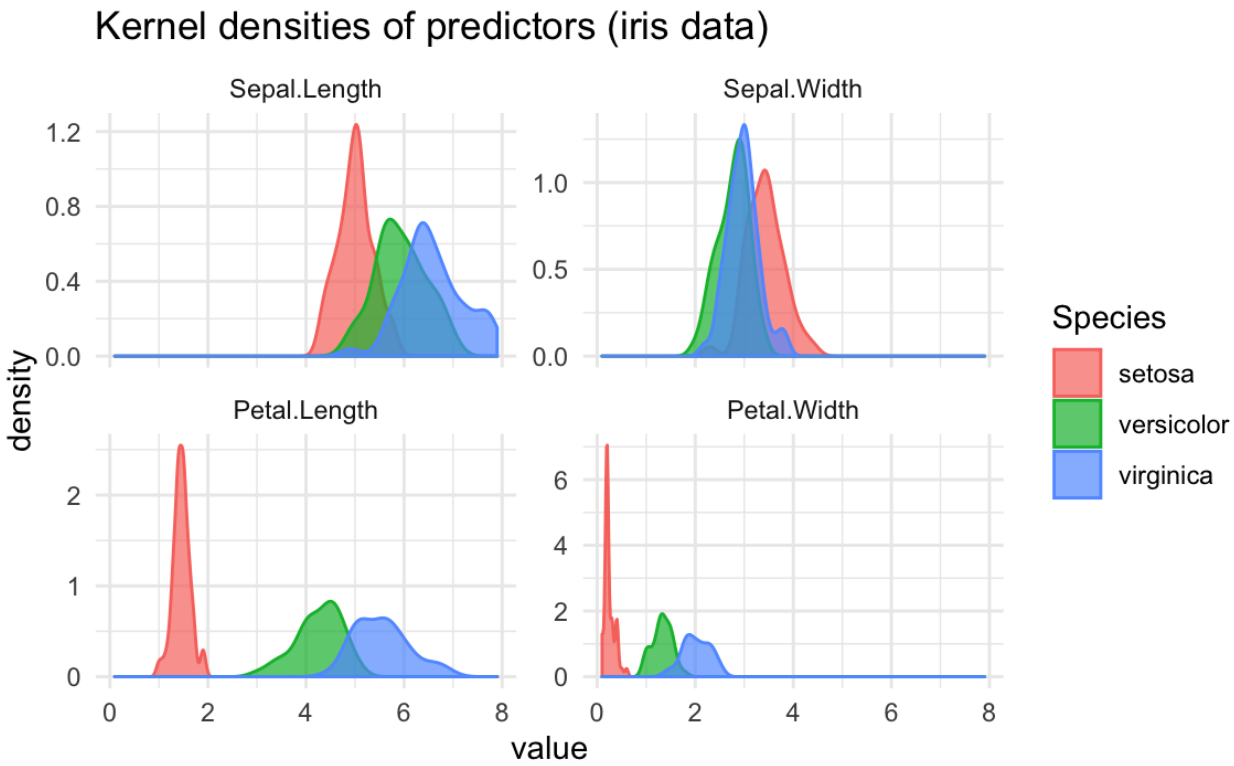
- The boxplots of some predictors are fairly different between iris species.



- The boxplots of petal length and petal width differ considerably between species.
- In contrast, predictors like sepal length and sepal width have more overlapping distributions across species.

Motivation

- The same differences can be seen for density curves.



Distinguishing Species

- Which predictor provides the "best" distinction between species?
- In a classification problem, the response variable Y provides a group or class structure to the data.
- We expect that the predictors will help us discriminate between one class and the others.
- The general idea is to look for systematic differences among classes.

Distinguishing Species

- Let's consider a single predictor X and a categorical response Y measured on n individuals.
- Assume there are K classes. Let C_k represent the k-th class in Y. Let n_k be the number of observations in class C_k. Then

$$n = n_1 + n_2 + \ldots + n_K = \sum_{k=1}^{K} n_k$$

- The global mean value of X is:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

- Each class k has its own mean $\bar{x}_k$:

$$\bar{x}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} x_{ik}$$

Distinguishing Species

- A measure of global dispersion in X is the total sum of squares:

$$\text{tss} = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

- Each class k has its own sum of squares ss_k:

$$\text{ss}_k = \sum_{i=1}^{n_k} (x_{ik} - \bar{x}_k)^2$$

- If there is no difference in the group means of X, then the means $\bar{x}_k$ should be similar. Otherwise, one or more of the mean values may differ.
- A useful measure to compare differences among the K means is the deviation from the overall mean: $\bar{x}_k - \bar{x}$.

Distinguishing Species

- To summarize the deviations of each group mean from the overall mean, the between-group sum of squares is:

$$\text{bss} = \sum_{k=1}^{K} n_k(\bar{x}_k - \bar{x})^2$$

- To summarize the dispersion within groups, the within-group sum of squares is:

$$\text{wss} = \sum_{k=1}^{K} \text{ss}_k = \sum_{k=1}^{K} \sum_{i=1}^{n_k} (x_{ik} - \bar{x}_k)^2$$

- We have three types of sums of squares:

$$\text{total} \quad \text{tss} = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$\text{between} \quad \text{bss} = \sum_{k=1}^{K} n_k(\bar{x}_k - \bar{x})^2$$

$$\text{within} \quad \text{wss} = \sum_{k=1}^{K} \sum_{i=1}^{n_k} (x_{ik} - \bar{x}_k)^2$$

Sum of Squares Decomposition

- We want to decompose the squared deviations $(x_{ik} - \bar{x})^2$ in terms of the class structure:

$$\begin{aligned} x_{ik} - \bar{x} &= x_{ik} - \bar{x}_k + \bar{x}_k - \bar{x} \\ &= (\bar{x}_k - \bar{x}) + (x_{ik} - \bar{x}_k) \end{aligned}$$

- We can decompose tss in terms of bss and wss as follows:

$$\underbrace{\sum_{k=1}^{K}\sum_{i=1}^{n_k}(x_{ik} - \bar{x})^2}_{tss} = \underbrace{\sum_{k=1}^{K} n_k(\bar{x}_k - \bar{x})^2}_{bss} + \underbrace{\sum_{k=1}^{K}\sum_{i=1}^{n_k}(x_{ik} - \bar{x}_k)^2}_{wss}$$

- In summary:

$$\text{tss} = \text{bss} + \text{wss}$$

Correlation Ratio

- The correlation ratio measures the proportion of total dispersion explained by differences between groups.
- The squared correlation ratio η^2 (proposed by Karl Pearson) is:

$$\eta^2(X, Y) = \frac{\text{bss}}{\text{tss}}$$

- η^2 takes values between 0 and 1.
- $\eta^2 = 0$ indicates that all group means are equal.
- $\eta^2 = 1$ indicates no dispersion within the respective groups.

F-Ratio

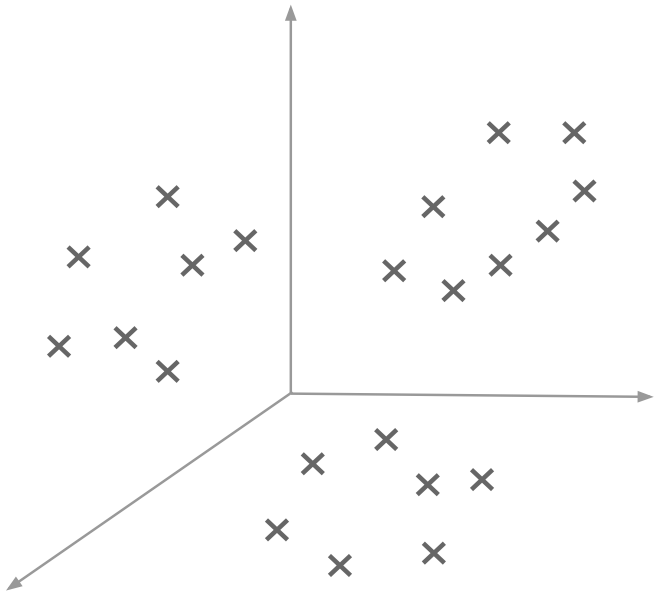
- The F-ratio (proposed by R.A. Fisher) is:

$$F = \frac{\text{bss}/(K-1)}{\text{wss}/(n-K)}$$

- The larger the value of both ratios (η^2 and F), the greater the dispersion between groups relative to the dispersion within groups.

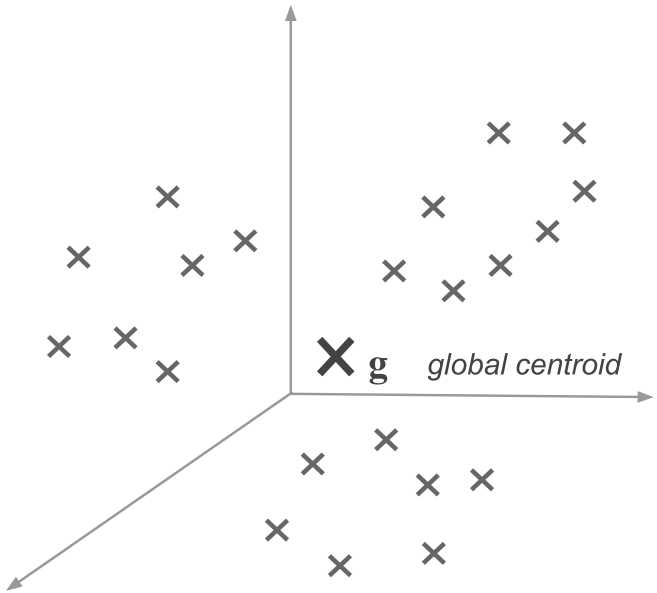
Geometric Perspective

- Assume that the objects form a cloud of points in p-dimensional space.



Geometric Perspective

- Consider the average individual $\mathbf{g}$, known as the global centroid (i.e., the center of gravity of the cloud of points):



- The global centroid is the point formed by the means of all variables: $\mathbf{g} = [\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_p]$, where

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^{n} x_{ij}$$

Geometric Perspective

- If all variables are mean-centered, the centroid is the origin $\mathbf{g} = [0, 0, \ldots, 0]$.
- Taking the global centroid as a point of reference, we compute the amount of spread or dispersion in the data.
- Assuming the centered features, a matrix of total dispersion is given by the Total Sums of Squares (TSS):

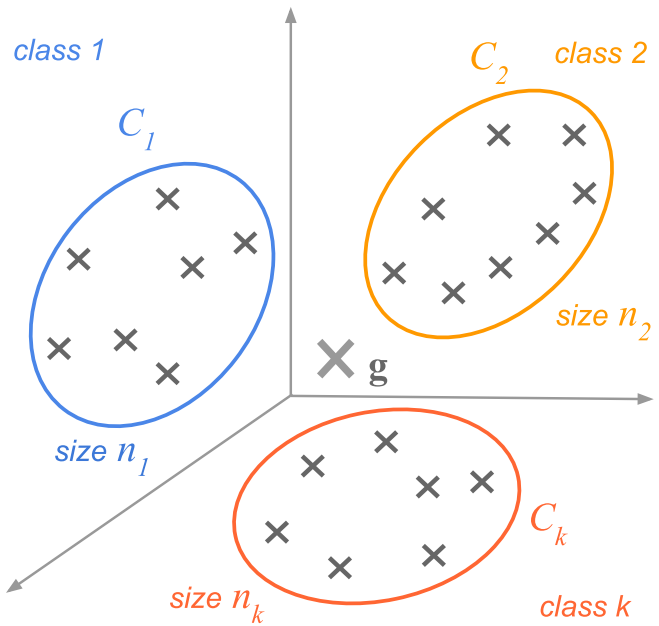
$$\text{TSS} = \mathbf{X}^\top \mathbf{X}$$

- Alternatively, we can get the sample variance-covariance matrix $\mathbf{V}$:

$$\mathbf{V} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X}$$

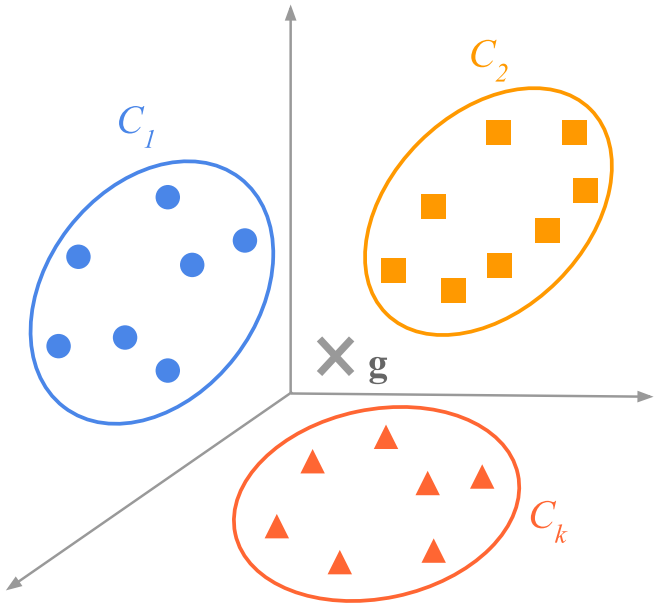
Geometric Perspective

- n_k: the number of observations in the k-th class.
- x_{ijk}: the value of the j-th variable for the i-th observation in the k-th class.
- $\mathbf{x}_{ik}$: the feature vector of the i-th observation in the k-th class.
- $\bar{x}_{jk}$: the mean of the j-th variable in the k-th class.

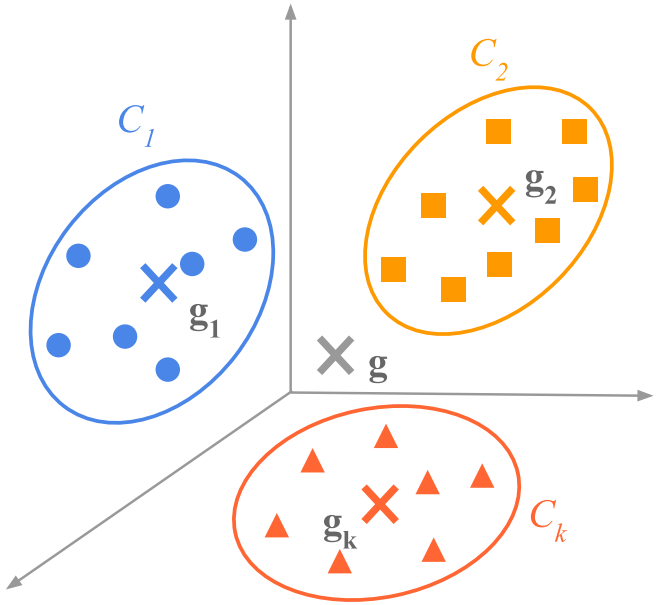


Geometric Perspective

Each class is denoted as C_k and is formed by n_k individuals.



Geometric Perspective

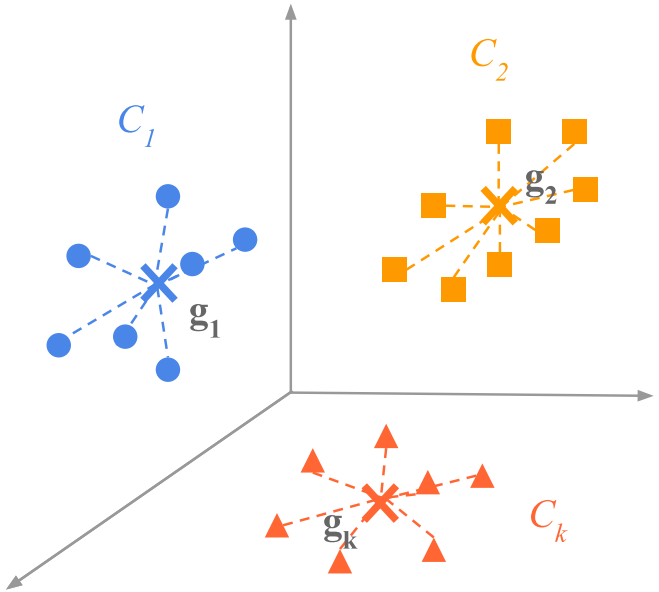


- We can look at the local or class centroids (one per class).
- The class centroid $\mathbf{g}_k$ is the point formed by the means of the observations in class k: $\mathbf{g}_k = [\bar{x}_{1k}, \bar{x}_{2k}, \ldots, \bar{x}_{pk}]$, where:

$$\bar{x}_{jk} = \frac{1}{n_k} \sum_{i=1}^{n_k} x_{ijk}$$

Geometric Perspective

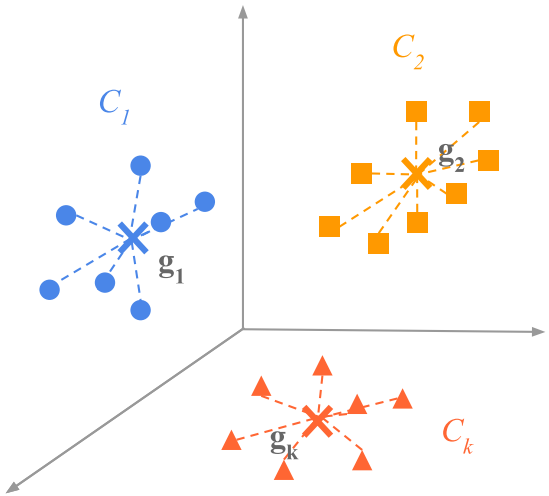
- We consider the dispersion within the clouds.



- Each group has an associated dispersion matrix, given by the Class Sum of Squares (CSS): $CSS_k = \mathbf{X}_k^\top \mathbf{X}_k$ where $\mathbf{X}_k$ is the data matrix centered around the class centroid.
- There is an associated variance-covariance matrix $\mathbf{W}_k$ for each class:

$$\mathbf{W}_k = \frac{1}{n_k - 1} \mathbf{X}_k^\top \mathbf{X}_k$$

Geometric Perspective



- We can combine the class dispersion to obtain a Within-class Sums of Squares (WSS) matrix:

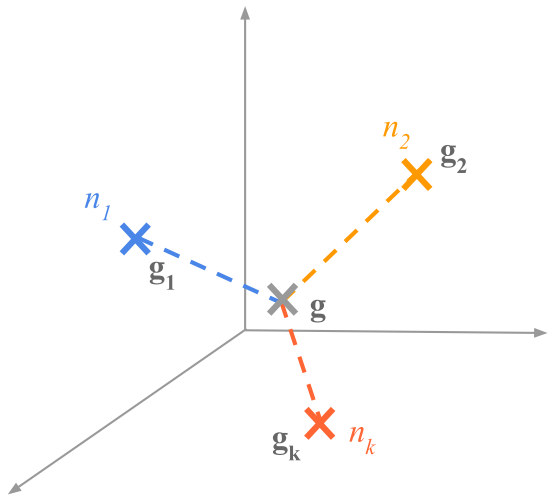
$$WSS = \sum_{k=1}^{K} CSS_k = \sum_{k=1}^{K} \mathbf{X}_k^\top \mathbf{X}_k$$

- We can combine the class covariance matrices $\mathbf{W}_k$ using weighted contributions to obtain the Within-class covariance matrix $\mathbf{W}$:

$$\mathbf{W} = \sum_{k=1}^{K} \frac{n_k - 1}{n - 1} \mathbf{W}_k$$

Geometric Perspective

- If we just focus on the centroids

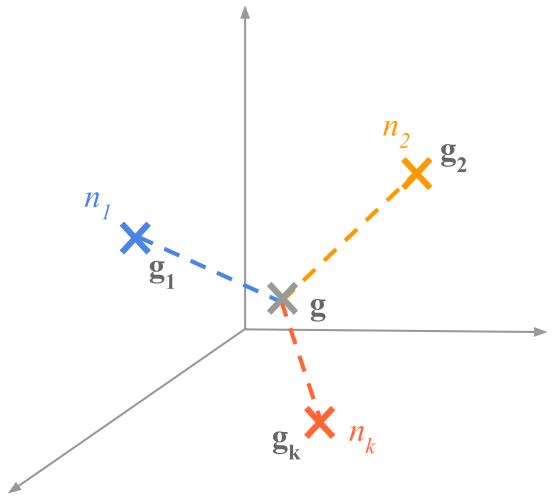


- The global centroid $\mathbf{g}$ can be expressed as a weighted average of the group centroids

$$\mathbf{g} = \frac{n_1}{n}\mathbf{g}_1 + \frac{n_2}{n}\mathbf{g}_2 + \cdots + \frac{n_K}{n}\mathbf{g}_K = \sum_{k=1}^{K} \frac{n_k}{n}\mathbf{g}_k$$

Geometric Perspective

- If we just focus on the centroids



- The matrix of dispersion Between Sum of Squares (BSS):

$$\text{BSS} = \sum_{k=1}^{K} n_k (\mathbf{g}_k - \mathbf{g})(\mathbf{g}_k - \mathbf{g})^\top$$

- The Between-Class Variance Matrix $\mathbf{B}$:

$$\mathbf{B} = \sum_{k=1}^{K} \frac{n_k}{n-1} (\mathbf{g}_k - \mathbf{g})(\mathbf{g}_k - \mathbf{g})^\top$$

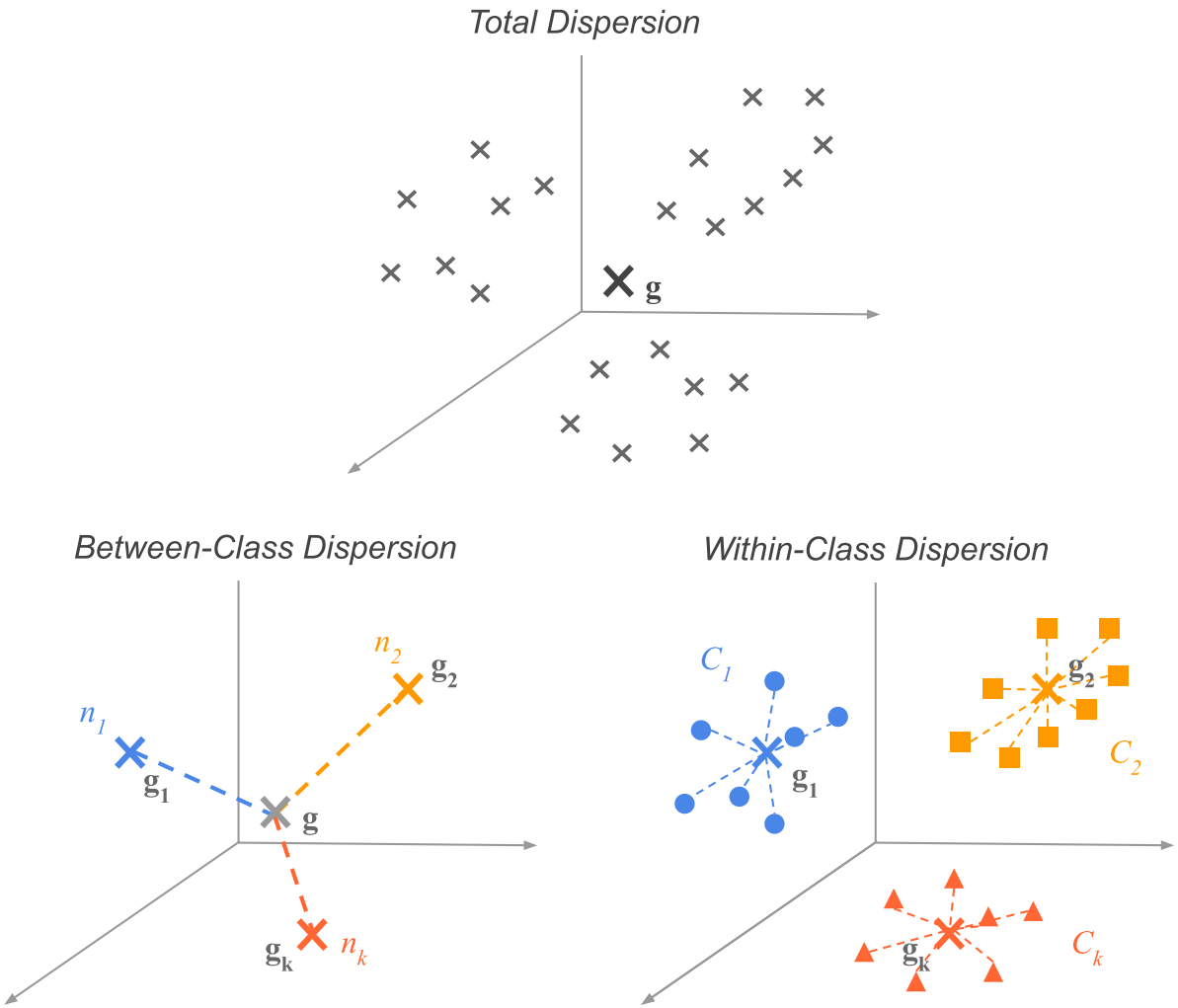
Geometric Perspective

- We have three types of Sum of Squares matrices:
  1. TSS: Total Sum of Squares
  2. WSS: Within-class Sum of Squares
  3. BSS: Between-class Sum of Squares
- We have three types of covariance matrices:
  1. $\mathbf{V}$: Total covariance
  2. $\mathbf{W}$: Within-class covariance contribution
  3. $\mathbf{B}$: Between-class covariance contribution

Dispersion Decomposition

- It can be shown that, for both sums of squares and covariance matrices, the total dispersion can be decomposed as:
  - TSS $=$ BSS $+$ WSS
  - $\mathbf{V} = \mathbf{B} + \mathbf{W}$
- Let $\mathbf{X}$ be the $n \times p$ mean-centered matrix of predictors and $\mathbf{Y}$ be the $n \times K$ class-indicator matrix, where $Y_{ik} = 1$ if observation i belongs to class k, and 0 otherwise.
  - TSS $= \mathbf{X}^\top \mathbf{X}$
  - BSS $= \mathbf{X}^\top \mathbf{Y} (\mathbf{Y}^\top \mathbf{Y})^{-1} \mathbf{Y}^\top \mathbf{X}$
  - WSS $= \mathbf{X}^\top (\mathbf{I} - \mathbf{Y} (\mathbf{Y}^\top \mathbf{Y})^{-1} \mathbf{Y}^\top) \mathbf{X}$

Dispersion Decomposition

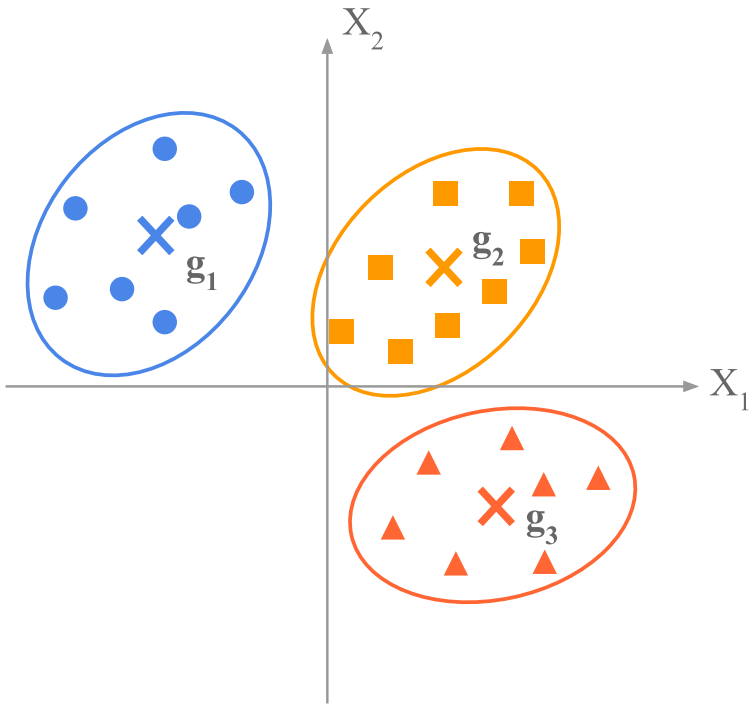


Canonical Discriminant Analysis (CDA)

- **Canonical Discriminant Analysis (CDA)** is closely related to Linear Discriminant Analysis (LDA).
- CDA can be considered from two perspectives:
  - Exploratory: How to find a representation of the objects which provides the best separation between classes?
  - Predictive: How to find the rules for assigning a class to a given object?

CDA: Exploratory Aspect

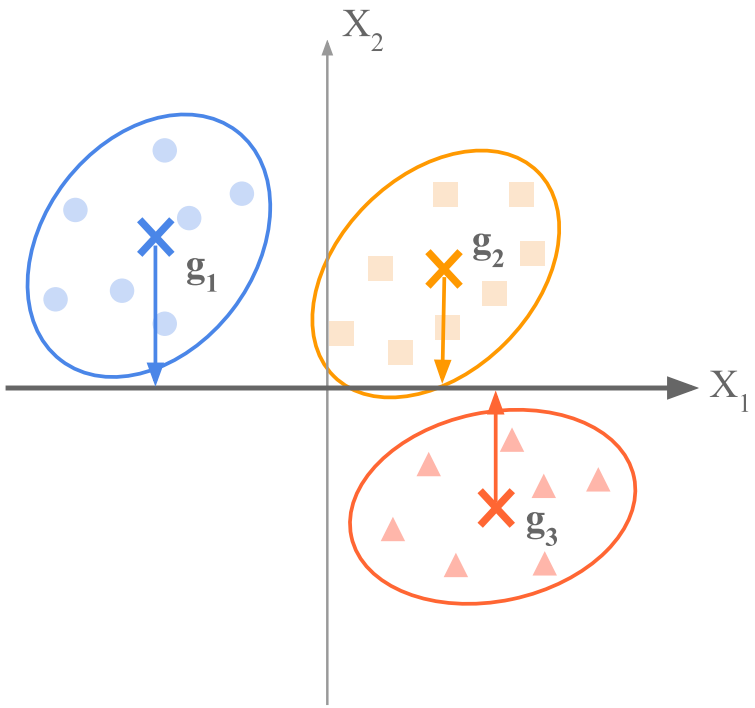
- We formulate the classification problem of CDA in a geometric way.
- For example, let's consider three classes in a 2D space as below.



- From an exploratory/descriptive perspective, we look for a good low dimensional representation that separates the three classes.

CDA: Exploratory Aspect

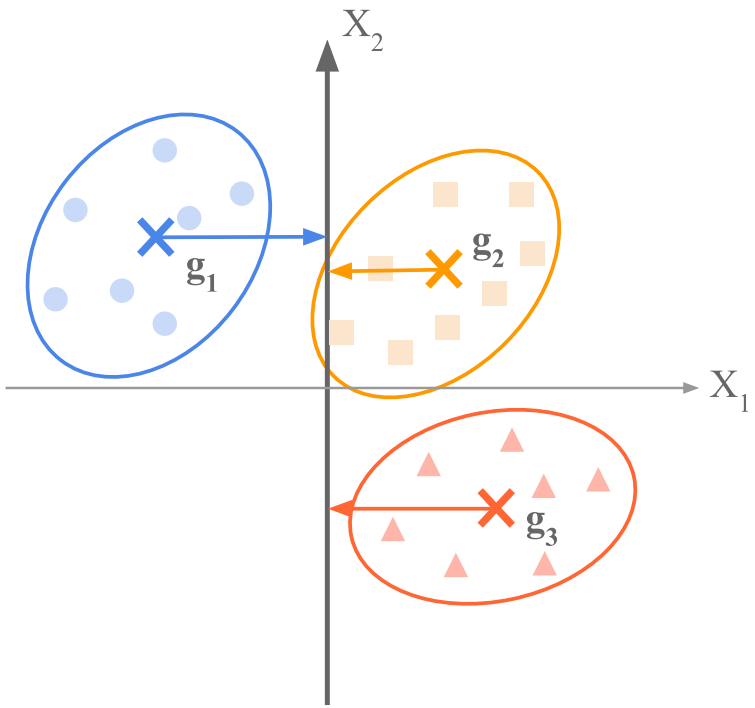
- We consider the axis associated with the predictor X_1, i.e., the horizontal axis.



- If we project the individuals on the X_1 axis, class 1 is separated from classes 2 and 3.
- However, class 2 is largely confounded with class 3.

CDA: Exploratory Aspect

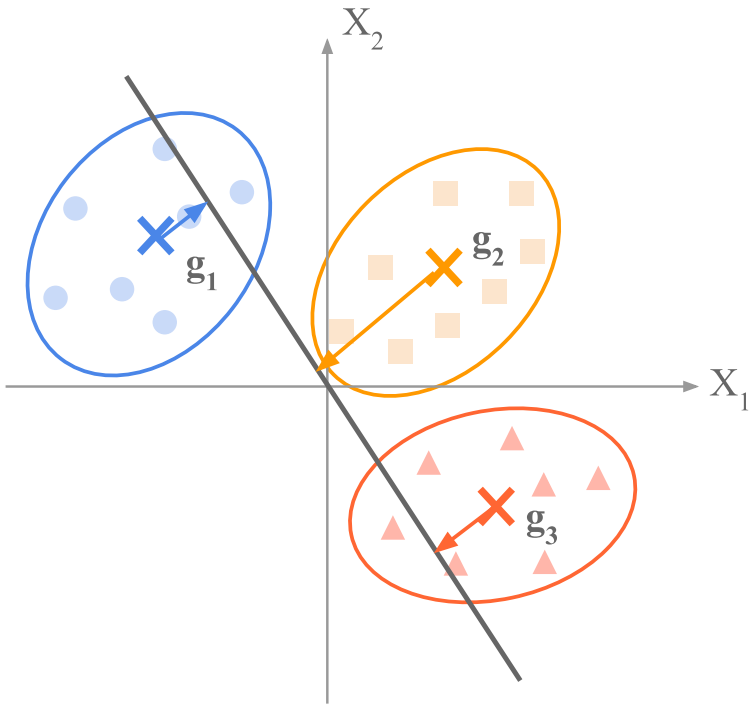
- We consider the axis associated with the predictor X_2, i.e., the vertical axis.



- If we project the individuals on the X_2 axis, class 3 is separated from classes 1 and 2. However, class 1 is largely confounded with class 2.
- Is there an axis that "best" separates the three clouds?

CDA: Exploratory Aspect

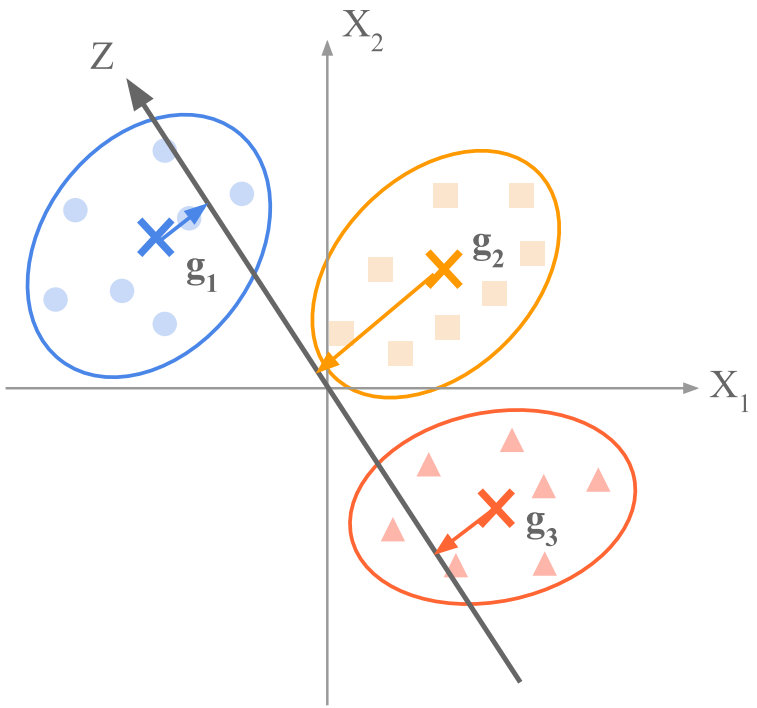
- We look for an optimal representation in the sense of finding an axis that best separates the clouds:



- The exploratory aspect of CDA involves seeking a low dimensional representation in which the class of objects are well separated.

Looking for a discriminant axis

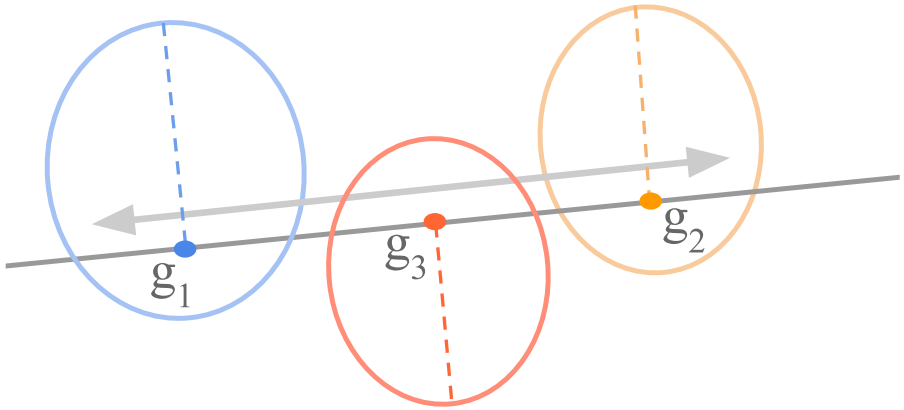
- We look for an axis Δ_u, spanned by some vector $\mathbf{u}$, that separates all three groups in an adequate way.



- We look for a linear combination of the predictors $\mathbf{z} = \mathbf{X}\mathbf{u}$ that *ideally* achieves the following goals:
  - Minimize within-class dispersion: $\min\{\mathbf{u}^\top \mathbf{W}\mathbf{u}\}$
  - Maximize between-class dispersion: $\max\{\mathbf{u}^\top \mathbf{B}\mathbf{u}\}$

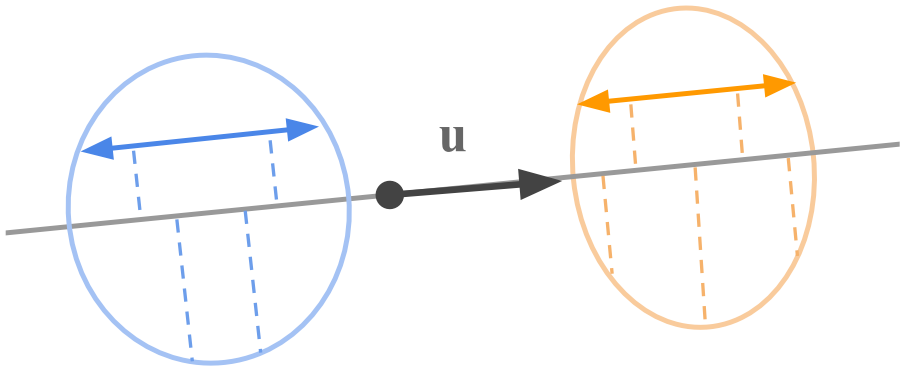
Looking for a discriminant axis

- We want to have $\mathbf{u}$ such that between class dispersion is maximized.
- This is a situation in which the class centroids are well separated:



Looking for a discriminant axis

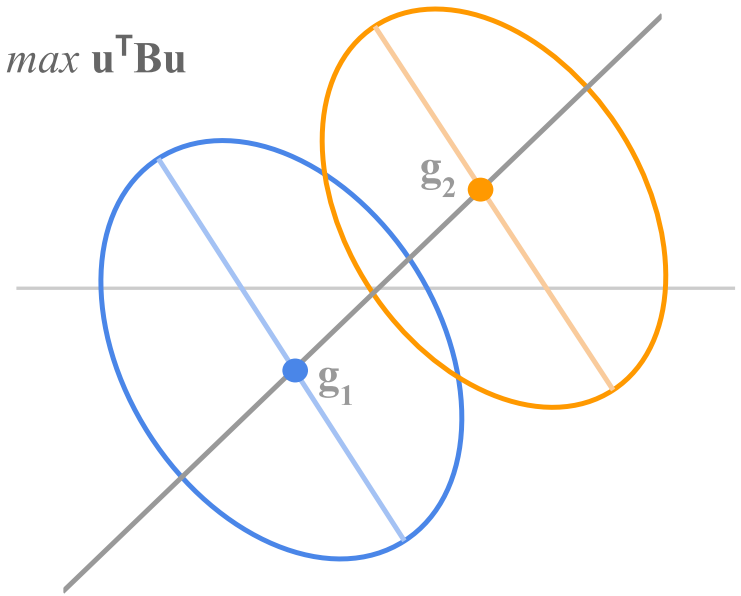
- We also want to have $\mathbf{u}$ such that the within-class dispersion is minimized.
- This implies having classes in which, on average, the "inner" variation is small (i.e., concentrated local dispersion).



- It is generally impossible to find an axis Δ_u, spanned by $\mathbf{u}$, which simultaneously maximizes the between-group variance and minimizes the within-group variance.

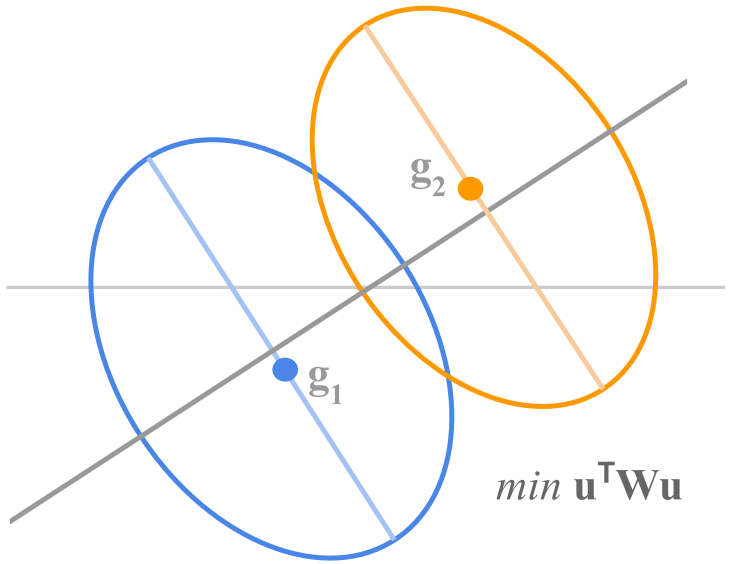
Looking for a discriminant axis

- For example, consider two classes as shown below. Maximizing the between-class separation involves choosing $\mathbf{u}_a$ parallel to the segment linking the centroids.
- The direction $\mathbf{u}_a$ is aligned with the two centroids.



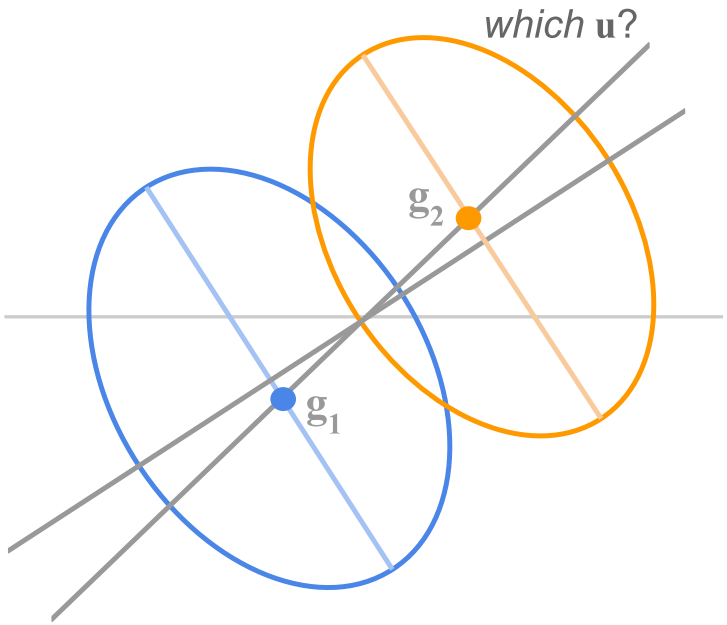
Looking for a discriminant axis

- Minimizing the within-class dispersion involves finding $\mathbf{u}_b$ perpendicular to the principal axis of the ellipses.
- This direction $\mathbf{u}_b$ does not necessarily pass through the centroids.



Looking for a discriminant axis

- In general, we end up with two possibilities $\mathbf{u}_a \neq \mathbf{u}_b$



Looking for a discriminant axis - A compromise criterion

- It is generally impossible to minimize $\mathbf{u}^\top \mathbf{W}\mathbf{u}$ and maximize $\mathbf{u}^\top \mathbf{B}\mathbf{u}$ simultaneously. We look for a compromise.
- Let's consider the variance decomposition.

$$\mathbf{V} = \mathbf{W} + \mathbf{B}$$

$$\mathbf{u}^\top \mathbf{V}\mathbf{u} = \underbrace{\mathbf{u}^\top \mathbf{W}\mathbf{u}}_{minimize} + \underbrace{\mathbf{u}^\top \mathbf{B}\mathbf{u}}_{maximize}$$

- We have two options:

$$\max\left\{\frac{\mathbf{u}^\top \mathbf{B}\mathbf{u}}{\mathbf{u}^\top \mathbf{V}\mathbf{u}}\right\} \qquad \max\left\{\frac{\mathbf{u}^\top \mathbf{B}\mathbf{u}}{\mathbf{u}^\top \mathbf{W}\mathbf{u}}\right\}$$

  which correspond to the correlation ratio η^2 and a criterion proportional to the F-ratio:

$$\eta^2 \longleftrightarrow \frac{\mathbf{u}^\top \mathbf{B}\mathbf{u}}{\mathbf{u}^\top \mathbf{V}\mathbf{u}} \qquad F \propto \frac{\mathbf{u}^\top \mathbf{B}\mathbf{u}}{\mathbf{u}^\top \mathbf{W}\mathbf{u}}$$

- Both criteria yield the same optimal discriminant directions.

Correlation Ratio Criterion

- Let's look for $\mathbf{u}$ such that $\max\left\{\frac{\mathbf{u}^\top\mathbf{B}\mathbf{u}}{\mathbf{u}^\top\mathbf{V}\mathbf{u}}\right\}$.
- For convenience, we use a normalization constraint $\mathbf{u}^\top\mathbf{V}\mathbf{u} = 1$.

$$\max\left\{\frac{\mathbf{u}^\top\mathbf{B}\mathbf{u}}{\mathbf{u}^\top\mathbf{V}\mathbf{u}}\right\} \Longleftrightarrow \max\{\mathbf{u}^\top\mathbf{B}\mathbf{u}\} \text{ s.t. } \mathbf{u}^\top\mathbf{V}\mathbf{u} = 1$$

- Using the method of Lagrange multipliers:

$$L(\mathbf{u}) = \mathbf{u}^\top\mathbf{B}\mathbf{u} - \lambda(\mathbf{u}^\top\mathbf{V}\mathbf{u} - 1)$$

- Differentiating w.r.t. $\mathbf{u}$ and equating to zero:

$$\frac{\partial L(\mathbf{u})}{\partial \mathbf{u}} = 2\mathbf{B}\mathbf{u} - 2\lambda\mathbf{V}\mathbf{u} = \mathbf{0}$$

$$\mathbf{B}\mathbf{u} = \lambda\mathbf{V}\mathbf{u}$$

$$\mathbf{V}^{-1}\mathbf{B}\mathbf{u} = \lambda\mathbf{u} \qquad \text{if } \mathbf{V} \text{ is invertible.}$$

- The optimal vector $\mathbf{u}$ is an eigenvector of $\mathbf{V}^{-1}\mathbf{B}$ associated with its largest eigenvalue.

F-Ratio Criterion

- Similarly, let's look for $\mathbf{u}$ such that:

$$\max\left\{\frac{\mathbf{u}^\top \mathbf{B}\mathbf{u}}{\mathbf{u}^\top \mathbf{W}\mathbf{u}}\right\} \Longleftrightarrow \max\{\mathbf{u}^\top \mathbf{B}\mathbf{u}\} \text{ s.t. } \mathbf{u}^\top \mathbf{W}\mathbf{u} = 1$$

- Using the same Lagrangian procedure with a multiplier ρ, we have:

$$\mathbf{B}\mathbf{u} = \rho\mathbf{W}\mathbf{u}$$

and if $\mathbf{W}$ is invertible, then $\mathbf{u}$ is an eigenvector of $\mathbf{W}^{-1}\mathbf{B}$ associated with eigenvalue ρ:

$$\mathbf{W}^{-1}\mathbf{B}\mathbf{u} = \rho\mathbf{u}$$

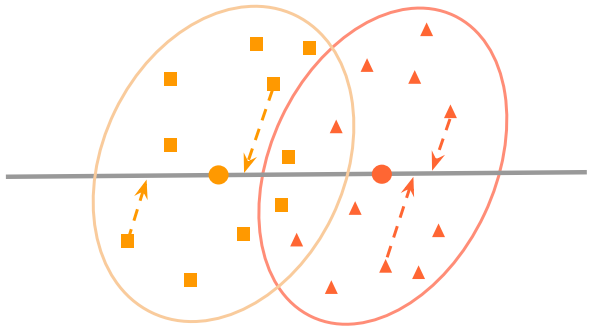
The optimal vector corresponds to the largest eigenvalue ρ.

- The relationship between the eigenvalues λ and ρ is:

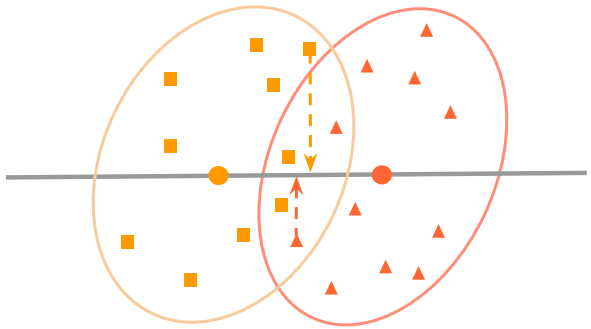
$$\rho = \frac{\lambda}{1-\lambda}$$

A Special PCA

- The vector $\mathbf{u}$ defines a discriminant axis related to PCA on the cloud of centroids $\mathbf{g}_1, \ldots, \mathbf{g}_K$, but using a different metric. The projection may be viewed as oblique in the original space.



- Without accounting for the covariance structure ($\mathbf{V}^{-1}$ or $\mathbf{W}^{-1}$), this would resemble ordinary PCA on the centroids, using orthogonal projections that may provide less class separation:



CDA: Predictive Aspect

- How do we use CDA for classification purposes?
- This involves establishing a decision rule that lets us predict the class of an object.

Distance behind CDA

- Consider two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^p$. The inner product of $\mathbf{a}$ and $\mathbf{b}$ is $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^\top \mathbf{b} = \mathbf{a}^\top \mathbf{I}_p \mathbf{b}$, where $\mathbf{I}_p$ denotes the $p \times p$ identity matrix.
- For a symmetric positive-definite matrix $\mathbf{M}$ (called the **metric matrix**), we define the inner product of $\mathbf{a}$ and $\mathbf{b}$, under metric $\mathbf{M}$, to be:

$$\langle \mathbf{a}, \mathbf{b} \rangle_{\mathbf{M}} = \mathbf{a}^\top \mathbf{M} \mathbf{b}$$

- The squared Euclidean distance between two vectors $\mathbf{x}_i$ and $\mathbf{x}_l$:

$$\begin{aligned} d^2(i,l) &= \langle \mathbf{x}_i - \mathbf{x}_l, \mathbf{x}_i - \mathbf{x}_l \rangle \\ &= (\mathbf{x}_i - \mathbf{x}_l)^\top (\mathbf{x}_i - \mathbf{x}_l) \\ &= (\mathbf{x}_i - \mathbf{x}_l)^\top \mathbf{I}_p (\mathbf{x}_i - \mathbf{x}_l) \end{aligned}$$

- Similarly, we replace $\mathbf{I}_p$ with a metric matrix $\mathbf{M}$ to obtain a generalized distance:

$$d^2_{\mathbf{M}}(i,l) = (\mathbf{x}_i - \mathbf{x}_l)^\top \mathbf{M} (\mathbf{x}_i - \mathbf{x}_l)$$

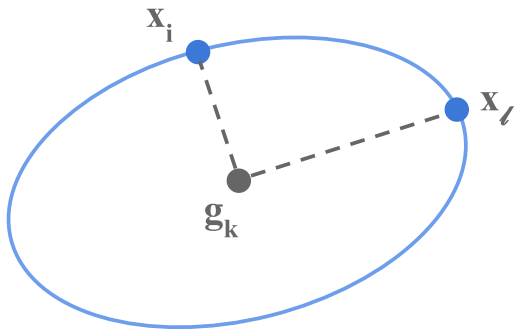
Distance behind CDA

- The classification rule used in CDA consists of assigning each individual $\mathbf{x}_i$ to the class C_k for which the distance to the centroid $\mathbf{g}_k$ is minimal.
- Instead of the Euclidean distance, we use the **Mahalanobis** distance, based on the metric matrix $\mathbf{W}^{-1}$ (assuming $\mathbf{W}$ is positive definite). The squared distance is:

$$\text{Mahalanobis:} \quad d^2(\mathbf{x}_i, \mathbf{g}_k) = (\mathbf{x}_i - \mathbf{g}_k)^\top \mathbf{W}^{-1} (\mathbf{x}_i - \mathbf{g}_k)$$

- The squared Mahalanobis distance measures the separation between a point $\mathbf{x}_i$ and a centroid $\mathbf{g}_k$, taking into account the covariance structure of the variables.

Distance behind CDA



Euclidean distance

$$d^2(\mathbf{x}_i, \mathbf{g}_k) \neq d^2(\mathbf{x}_\ell, \mathbf{g}_k)$$

$\mathbf{x}_i$ and $\mathbf{x}_\ell$ are at different distances to centroid $\mathbf{g}_k$

Mahalanobis distance

$$d^2(\mathbf{x}_i, \mathbf{g}_k) = d^2(\mathbf{x}_\ell, \mathbf{g}_k)$$

$\mathbf{x}_i$ and $\mathbf{x}_\ell$ are equidistant to centroid $\mathbf{g}_k$

- The distance of a point $\mathbf{x}_i$ to the centroid $\mathbf{g}_k$ depends on the spread and covariance structure of the data.
- If two points i and l lie on the same Mahalanobis-distance ellipsoid, they are equidistant from the centroid:

$$d^2_{\mathbf{W}^{-1}}(\mathbf{x}_i, \mathbf{g}_k) = d^2_{\mathbf{W}^{-1}}(\mathbf{x}_l, \mathbf{g}_k)$$

CDA Classifier

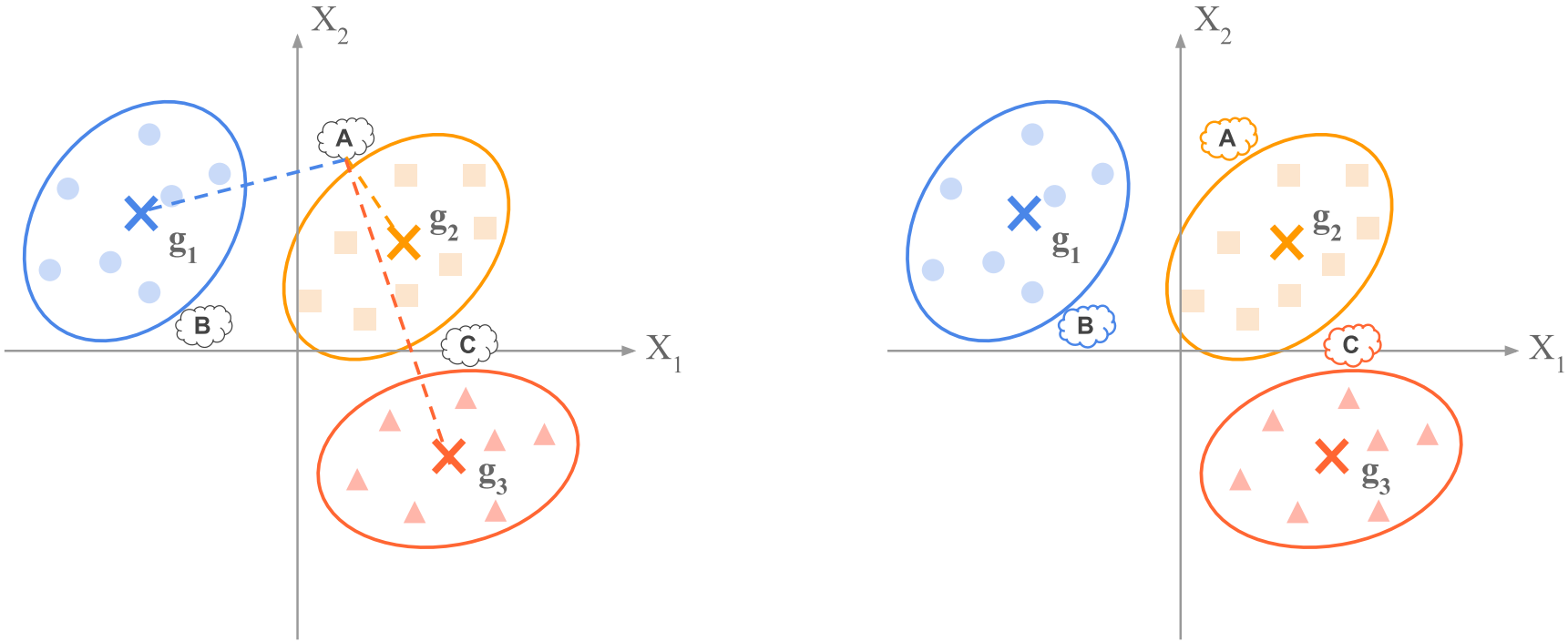
- The classification rule behind CDA is that we should assign an object $\mathbf{x}_i$ to the class it is nearest to, using the $\mathbf{W}^{-1}$ metric matrix to calculate the distance of the object from the centroid of the group.
- The Mahalanobis distance of $\mathbf{x}_i$ to centroid $\mathbf{g}_k$:

$$\begin{aligned} d^2(\mathbf{x}_i, \mathbf{g}_k) &= (\mathbf{x}_i - \mathbf{g}_k)^\top \mathbf{W}^{-1} (\mathbf{x}_i - \mathbf{g}_k) \\ &= \underbrace{\mathbf{x}_i^\top \mathbf{W}^{-1} \mathbf{x}_i}_{constant} - \underbrace{2\mathbf{g}_k^\top \mathbf{W}^{-1} \mathbf{x}_i}_{\text{depends on } k} + \underbrace{\mathbf{g}_k^\top \mathbf{W}^{-1} \mathbf{g}_k}_{\text{depends on } k} \end{aligned}$$

- Minimizing $d^2(\mathbf{x}_i, \mathbf{g}_k)$ is equivalent to maximizing:

$$2\mathbf{g}_k^\top \mathbf{W}^{-1} \mathbf{x}_i - \mathbf{g}_k^\top \mathbf{W}^{-1} \mathbf{g}_k$$

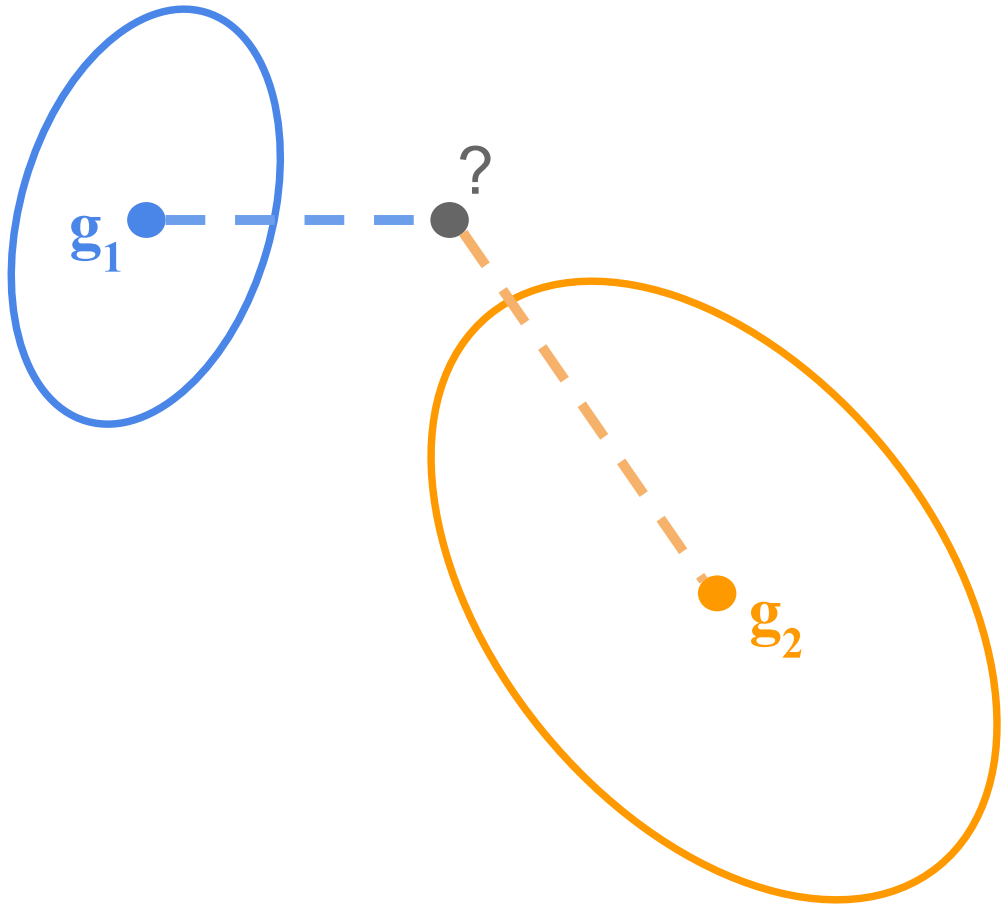
CDA Classifier



- For example, we have three unclassified individuals A, B, and C.
- We compute the Mahalanobis distances of the unclassified objects to the three centroids.
- With individual A, for example, we have to compute $d^2(\mathbf{x}_A, \mathbf{g}_1), d^2(\mathbf{x}_A, \mathbf{g}_3), d^2(\mathbf{x}_A, \mathbf{g}_2)$ and select class C_2 for which the Mahalanobis distance is minimal.

Limitations of CDA Classifier

- The CDA geometric classification rule may not be appropriate when classes have different *a priori* probabilities or different covariance matrices.



Probabilistic Discriminant Analysis

- Suppose we have a response with K classes $C_1, C_2, \ldots, C_K$ and p predictors. The Bayes classifier gives the classification rule:

  assign $\mathbf{x}_i$ to the class for which $P(y_i = k \mid \mathbf{x}_i)$ is the largest

- The rule is optimal because it minimizes the expected classification error under 0–1 loss.

$$\underbrace{P(y_i = k \mid \mathbf{x}_i)}_{\text{posterior}} = \frac{\overbrace{f_k(\mathbf{x}_i)}^{\text{likelihood}} \overbrace{P(y_i = k)}^{\text{prior}}}{f(\mathbf{x}_i)}$$

where the denominator is obtained as:

$$f(\mathbf{x}_i) = \sum_{k=1}^{K} P(y_i = k) f_k(\mathbf{x}_i)$$

Probabilistic Discriminant Analysis

$$\underbrace{P(y_i = k \mid \mathbf{x}_i)}_{\text{posterior}} = \frac{\overbrace{f_k(\mathbf{x}_i)}^{\text{likelihood}} \overbrace{P(y_i = k)}^{\text{prior}}}{f(\mathbf{x}_i)}$$

- Changing the notation, let:
  - $P(y = k) = \pi_k$: the prior probability of class k.
  - $f_k(\mathbf{x})$: the class-conditional density of the predictors given class k.
- Thus, the posterior probability (the conditional probability of the response given the inputs) is:

$$P(y = k \mid X = \mathbf{x}) = \frac{f_k(\mathbf{x})\pi_k}{\sum_{j=1}^{K} f_j(\mathbf{x})\pi_j}$$

The denominator is the same for all classes $k = 1, \ldots, K$.

Normal Distributions

- **We assume that our class-conditional densities follow Gaussian distributions.**
- For univariate data ($p = 1$), the class-conditional density has a Normal distribution $x|C_k \sim N(\mu_k, \sigma_k^2)$:

$$f_k(x) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2\right\}$$

- For multivariate data ($p > 1$), $\mathbf{x}|C_k \sim N(\mu_k, \mathbf{\Sigma}_k)$:

$$f_k(\mathbf{x}) = \frac{1}{(2\pi)^{p/2}|\mathbf{\Sigma}_k|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\mu_k)^\top \mathbf{\Sigma}_k^{-1}(\mathbf{x}-\mu_k)\right\}$$

where $\mathbf{\Sigma}_k$ and μ_k are the covariance matrix and the mean vector of class C_k. The exponent is minus one-half of the squared Mahalanobis distance:

$$d^2(\mathbf{x}, \mu_k) = (\mathbf{x}-\mu_k)^\top \mathbf{\Sigma}_k^{-1}(\mathbf{x}-\mu_k)$$

Estimating Parameters of Normal Distributions

- To use the previous equations, we need to estimate: prior probabilities $\hat{\pi}_k$, mean vectors $\hat{\mu}_k$, and covariance matrices $\hat{\mathbf{\Sigma}}_k$.
- Estimating π_k is straightforward:

$$\hat{\pi}_k = \frac{n_k}{n}$$

where $n_k = |C_k|$ is the size of class k and n is the total number of data points.

- For the mean vectors $\hat{\mu}_k$, we can use the centroid of class k: $\hat{\mu}_k = \mathbf{g}_k$.
- For the covariance matrices $\hat{\mathbf{\Sigma}}_k$, we can use the sample covariance matrix of each class:

$$\hat{\mathbf{\Sigma}}_k = \frac{1}{n_k - 1}\mathbf{X}_k^\top \mathbf{X}_k$$

where $\mathbf{X}_k$ is the data matrix centered around the mean of class k.

Discriminant Functions

- Given all the estimates, we can now estimate the posterior probability $P(y_i = k \mid \mathbf{x}_i)$:

$$\hat{P}(y_i = k \mid \mathbf{x}) = \frac{\hat{\pi}_k \hat{f}_k(\mathbf{x})}{\sum_{j=1}^{K} \hat{\pi}_j \hat{f}_j(\mathbf{x})}$$

- Because the denominator remains constant across classes, we can focus on the numerator:

$$\arg\max_k \hat{P}(y_i = k \mid \mathbf{x}) = \arg\max_k \left[\ln \hat{\pi}_k + \ln \hat{f}_k(\mathbf{x})\right]$$

- Considering the multivariate Normal density, we have:

$$\ln\left[\hat{f}_k(\mathbf{x})\right] = \ln\left[(2\pi)^{-p/2}|\hat{\mathbf{\Sigma}}_k|^{-1/2}\exp\left\{-\frac{1}{2}(\mathbf{x}-\hat{\mu}_k)^\top \hat{\mathbf{\Sigma}}_k^{-1}(\mathbf{x}-\hat{\mu}_k)\right\}\right]$$

$$\longrightarrow -\frac{1}{2}\ln\left(|\hat{\mathbf{\Sigma}}_k|\right) - \frac{1}{2}\left[\mathbf{x}^\top \hat{\mathbf{\Sigma}}_k^{-1}\mathbf{x} - 2\mathbf{x}^\top \hat{\mathbf{\Sigma}}_k^{-1}\hat{\mu}_k + \hat{\mu}_k^\top \hat{\mathbf{\Sigma}}_k^{-1}\hat{\mu}_k\right]$$

Discriminant Functions

- Substituting the previous expression into $\ln P(y = k \mid \mathbf{x})$, we obtain:

$$\ln P(y = k \mid \mathbf{x}) = -\frac{1}{2}\ln|\hat{\mathbf{\Sigma}}_k| - \frac{1}{2}\Big[\mathbf{x}^\top \hat{\mathbf{\Sigma}}_k^{-1}\mathbf{x} - 2\mathbf{x}^\top \hat{\mathbf{\Sigma}}_k^{-1}\hat{\mu}_k + \hat{\mu}_k^\top \hat{\mathbf{\Sigma}}_k^{-1}\hat{\mu}_k\Big] + \ln \hat{\pi}_k + \text{constant}$$

- Ignoring terms independent of k, we obtain the discriminant functions:

$$\delta_k(\mathbf{x}) = \ln \hat{\pi}_k - \frac{1}{2}\ln|\hat{\mathbf{\Sigma}}_k| - \frac{1}{2}(\mathbf{x} - \hat{\mu}_k)^\top \hat{\mathbf{\Sigma}}_k^{-1}(\mathbf{x} - \hat{\mu}_k)$$

- Our classification rule is then:

assign $\mathbf{x}$ to the class for which $\delta_k(\mathbf{x})$ is the largest.

Quadratic Discriminant Analysis (QDA)

- We classify the order of each term in $\delta_k(\mathbf{x})$ with respect to $\mathbf{x}$:

$$\underbrace{-\frac{1}{2}\ln\left(|\hat{\mathbf{\Sigma}}_k|\right)}_{\text{class-dependent}} - \frac{1}{2}\left[\underbrace{\mathbf{x}^\top\hat{\mathbf{\Sigma}}_k^{-1}\mathbf{x}}_{\text{quadratic}} - \underbrace{2\mathbf{x}^\top\hat{\mathbf{\Sigma}}_k^{-1}\hat{\mu}_k}_{\text{linear}} + \underbrace{\hat{\mu}_k^\top\hat{\mathbf{\Sigma}}_k^{-1}\hat{\mu}_k}_{\text{constant}}\right] + \underbrace{\ln\left(\hat{\pi}_k\right)}_{\text{class-dependent}}$$

- We have a quadratic function of $\mathbf{x}$. This leads us to Quadratic Discriminant Analysis (QDA).
- Having a quadratic discriminant function causes the decision boundaries in QDA to be quadratic surfaces.

Linear Discriminant Analysis (LDA)

- Assume that all classes have the same covariance matrix:

$$\hat{\boldsymbol{\Sigma}}_1 = \hat{\boldsymbol{\Sigma}}_2 = \ldots = \hat{\boldsymbol{\Sigma}}_K = \hat{\boldsymbol{\Sigma}}$$

- The discriminant function $\delta_k(\mathbf{x})$ becomes:

$$\underbrace{-\frac{1}{2}\ln\left(|\hat{\boldsymbol{\Sigma}}|\right)}_{\text{no } k \text{ dependency}} -\frac{1}{2}\Bigg[\underbrace{\mathbf{x}^\top\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{x}}_{\text{no } k \text{ dependency}} - \underbrace{2\mathbf{x}^\top\hat{\boldsymbol{\Sigma}}^{-1}\hat{\mu}_k}_{\text{linear}} + \underbrace{\hat{\mu}_k^\top\hat{\boldsymbol{\Sigma}}^{-1}\hat{\mu}_k}_{\text{constant}}\Bigg] + \underbrace{\ln\left(\hat{\pi}_k\right)}_{\text{class-dependent}}$$

- Ignoring the terms that do not depend on k, we obtain:

$$\delta_k(\mathbf{x}) = -\frac{1}{2}\Bigg[\underbrace{-2\mathbf{x}^\top\hat{\boldsymbol{\Sigma}}^{-1}\hat{\mu}_k}_{\text{linear}} + \underbrace{\hat{\mu}_k^\top\hat{\boldsymbol{\Sigma}}^{-1}\hat{\mu}_k}_{\text{constant}}\Bigg] + \underbrace{\ln\left(\hat{\pi}_k\right)}_{\text{class-dependent}}$$

- The discriminant function is now linear w.r.t. $\mathbf{x}$, giving Linear Discriminant Analysis (LDA).

Canonical Discriminant Analysis (CDA)

- We again assume that all covariance matrices are the same across classes, as well as the prior probabilities:

$$\hat{\boldsymbol{\Sigma}}_1 = \hat{\boldsymbol{\Sigma}}_2 = \ldots = \hat{\boldsymbol{\Sigma}}_K = \hat{\boldsymbol{\Sigma}} \quad \text{and} \quad \hat{\pi}_1 = \hat{\pi}_2 = \ldots = \hat{\pi}_K = \hat{\pi}$$

- The discriminant function $\delta_k(\mathbf{x})$ becomes:

$$\delta_k(\mathbf{x}) = -\frac{1}{2}\left[\underbrace{-2\mathbf{x}^\top \hat{\boldsymbol{\Sigma}}^{-1}\hat{\mu}_k}_{\text{linear}} + \underbrace{\hat{\mu}_k^\top \hat{\boldsymbol{\Sigma}}^{-1}\hat{\mu}_k}_{\text{constant in } \mathbf{x}}\right] + \underbrace{\ln(\hat{\pi})}_{\text{no } k \text{ dependency}}$$

- After ignoring the k-independent terms, we obtain:

$$\delta_k(\mathbf{x}) = \underbrace{\mathbf{x}^\top \hat{\boldsymbol{\Sigma}}^{-1}\hat{\mu}_k}_{\text{linear}} - \frac{1}{2}\underbrace{\hat{\mu}_k^\top \hat{\boldsymbol{\Sigma}}^{-1}\hat{\mu}_k}_{\text{constant in } \mathbf{x}}$$

- Maximizing this function is equivalent to minimizing the squared Mahalanobis distance between $\mathbf{x}$ and the centroid of C_k. This gives the CDA classification rule.

Naive Bayes

- Another simplifying assumption is that the predictors are conditionally independent given the class. For Gaussian data, this means each class has a diagonal covariance matrix:

$$\hat{\mathbf{\Sigma}}_k = \begin{pmatrix} Var(X_1 \mid C_k) & 0 & \dots & 0 \\ 0 & Var(X_2 \mid C_k) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & Var(X_p \mid C_k) \end{pmatrix}$$

- Unlike LDA, Gaussian Naive Bayes allows different diagonal covariance matrices for different classes.
- The discriminant function with diagonal $\hat{\mathbf{\Sigma}}_k$ is:

$$\delta_k(\mathbf{x}) = -\frac{1}{2} \sum_{j=1}^{p} \left[\ln \hat{\sigma}_{jk}^2 + \frac{(x_j - \hat{\mu}_{jk})^2}{\hat{\sigma}_{jk}^2} \right] + \ln \hat{\pi}_k$$

- This gives the Gaussian version of Naive Bayes.

Comparisons

Method	Covariance Matrices	Priors	Decision Boundary
QDA	Class-specific	Arbitrary	Quadratic
LDA	Equal across classes	Arbitrary	Linear
CDA (classifier)	Equal across classes	Equal across classes	Linear
Gaussian Naive Bayes	Class-specific, diagonal	Arbitrary	Generally quadratic